DoangJoo "Alan" Synn

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Research Interests

My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

Education

Georgia Institute of Technology Aug. 2022–Present
Ph.D. Candidate in Computer Science

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University Mar. 2020–Feb. 2022
M.Eng. in Electrical and Computer Engineering

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*
- Advised by Prof. Jong-Kook Kim

Korea University Mar. 2012–Feb. 2020
B.Eng. in Electrical Engineering

- Alternative military service as a Software Engineer, 2016–2019

Selected Research Projects

MotionSmith CHI 2026
Sketch-based design of mechanical motion and fabricable artifacts

- Developed a computational design system that connects sketched motion paths, mechanism generation, interactive refinement, and fabrication-ready output.
- Developed through a year-long participatory design process with three expert automata makers; supported motion exploration and the fabrication of working mechanical artifacts.
- Publication C1; project page and source code.

MotionSmith for Young Makers Manuscript
Educator co-design and classroom enactment of a mechanical CAD system

- Extended the research to classroom use through co-design with four K–8 STEM educators, revising the authoring workflow, reusable fabrication kit, and instructional supports.
- Used in educator-led classroom activities with 129 middle-school students at two U.S. schools; three educators facilitated STEM, STEAM, and engineering activities over three to four sessions per class.
- Manuscript M1.

Publications

Refereed Conference Papers

C1 **DoangJoo Synn** and Jong-Kook Kim. "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders." *Proceedings of ACK 2021*, 2021. doi:10.3745/PKIPS.y2021m11a.446.

C2 Byung Hoon Ahn, **DoangJoo Synn**, Masih Derkani, Eiman Ebrahimi, and Hadi Esmaeilzadeh. "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection." *Neural Information Processing Systems Demonstrations*, 2021.

C3 **DoangJoo Synn**, Zhifan Guo, Sehoon Ha, and HyunJoo Oh. "MotionSmith: A Sketch-Based Design System for Automata Making." *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi:10.1145/3772318.3791545.

C4 Jin Yu, Poojita Garg, **DoangJoo Synn**, and HyunJoo Oh. "Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning." *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025. doi:10.1145/3706598.3713260.

C5 JooYoung Park, **DoangJoo Synn**, and Jong-Kook Kim. "A Survey on the Advancement of Virtualization Technology." *Proceedings of Korea Information Processing Society 2021*, 2021. doi:10.3745/PKIPS.y2021m05a.12.

C6 DongJae Kim, **DoangJoo Synn**, and Jong-Kook Kim. "SG-Drop: Faster Skip-Gram by Dropping Context Words." *Proceedings of Korea Information Processing Society 2020*, 2020.

Refereed Journal Articles

J1 MyungHoon Ryu, XinYu Piao, JooYoung Park, **DoangJoo Synn**, and Jong-Kook Kim. "ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms." *The Transactions of the Korea Information Processing Society*, 2025. doi:10.3745/TKIPS.2025.14.1.1.

J2 **DoangJoo Synn**, XinYu Piao, JooYoung Park, and Jong-Kook Kim. "Distributed Training of Deep Learning Models Using Micro Batch Streaming." *KIPS Transactions on Software and Data Engineering*, 2023. doi:10.3745/KTSDE.2023.12.3.101.

J3 XinYu Piao, **DoangJoo Synn**, JooYoung Park, and Jong-Kook Kim. "Enabling large batch size training for dnn models beyond the memory limit while maintaining performance." *IEEE Access*, 2023. doi:10.1109/ACCESS.2023.3312572.

J4 Jung-Woo Han, **DoangJoo Synn**, Tae-Hyeong Kim, Hae-Cheon Jung, and Jong-Kook Kim. "Feature Based Sampling: A Fast and Robust Sampling Method for Tasks Using 3D Point Cloud." *IEEE Access*, 2022. doi:10.1109/ACCESS.2022.3178519.

Preprints

P1 JooYoung Park, **DoangJoo Synn**, XinYu Piao, and Jong-Kook Kim. "Dataloader Parameter Tuner: An Automated Dataloader Parameter Tuner for Deep Learning Models." *arXiv preprint arXiv:2210.05244*, 2022.

P2 **DoangJoo Synn**, Yuhan Wang, Sehoon Ha, and HyunJoo Oh. "MotionSmith for Young Makers: Educator Co-Design and Enactment of a Mechanical CAD System." 2027.

Patent Applications

IP1 HyunJoo Oh, Jin Yu, **DoangJoo Synn**, and Poojita Garg. "Coding Blocks Learning System and Method." Published U.S. patent application US 2026/0112292 A1, 23 April 2026. Application No. 19/357,594; filed 14 October 2025. Applicant: Georgia Tech Research Corporation.

Professional Experience

Dolby Laboratories May 2024–Aug. 2025
Ph.D. Research Intern, Experience Delivery Lab

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

LG Uplus June–Aug. 2023
MLOps Engineering Intern

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Protopia AI Jan. 2021–Aug. 2022
Research Scientist
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- Led framework development and Kubernetes/GPU infrastructure integration; the work contributed to a NeurIPS 2021 Demo.

Cloud Architect · AWS Educate Cloud Ambassador 2020–2022
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.
- Built cloud-native infrastructure using Kubernetes, Terraform, CI/CD, observability, and cost-optimized autoscaling while mentoring early-career engineers.

Flysher Jan.–Dec. 2019
Server Engineer
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.
- Built an ETL-to-Redshift analytics pipeline supporting data-informed feature rollout.

AKA Intelligence Apr. 2017–Jan. 2019
System Engineer
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.
- Coordinated hardware-software integration and production QA with overseas ODMs.

Teaching

Georgia Institute of Technology Aug. 2024–Present
Head Teaching Assistant, CS 4496/7496: Computer Animation
- School of Interactive Computing.

Korea University Science & Technology School Feb. 2021; Jan. 2022
Lecturer and Program Developer
- Taught introductory Java programming to 10+ undergraduate students; contributed to the BK21 Socialware IT education program.

Korea University Spring; Fall 2021
Teaching Assistant
- Digital System Design
- Data Structures and Algorithms

KAIST Engineering School Jan. 2012–Aug. 2014
Lecturer and Curriculum Developer
- Co-developed national project-based coding education programs with South Korea's Ministry of Science and ICT.
- Designed and evaluated 20+ curricula combining Arduino, hardware prototyping, and software development for high-school and college students.

Invited Talks & Outreach

Invited Panelist and MotionSmith Demonstrator, AutomataCon 2026
Motion, Making & Mechanical Design — GIFT Workshop, Georgia Tech CEISMC 2026
Invited Talk, Korea University 2026
MotionSmith Presentation, CHI 2026

Scholarly Activities

Reviewer, CHI 2027
Reviewer, CHI 2026
Subreviewer, CHI 2025
Artifact Evaluation Committee, ASPLOS 2024 2024

Honors & Awards

Fellowships
- Brain Korea 21 (BK21) Research Fellowship (2020–2022)

Awards
- Prize Winner, Creative Challenge Program @ Korea Univ. CTL (2015 and 2016)

Activities & Service

Georgia Tech Korean Student Association 2023–2024
- President of Korean Students in the College of Computing.

Hardware & Software Community, Korea University 2014–2019
- President; grew the community from 10 to 200+ active members across hardware and software tracks.

TEDxKoreaUniversity, Korea University 2015–2016
- Founder & Chief Organizer; led a 47-person team in producing a 200-guest event with 7 speakers and a $4K budget.